

Feature inventory for a new AP product

Build brief — PDF/image ingestion via email and shared folders

Saravanan Sadasivan · Digital Product Manager · v1.0

Purpose and scope

This is a consolidated feature inventory drawn from Capture (OCR) and AP Advance (Process), plus the reimagined now/next/future roadmap. Use it as a build brief to scope a new AP product with Claude. Every feature is something already specified or shipped somewhere in the suite, so the list is buildable, not aspirational.

In scope: Invoice ingestion as PDF and image only, arriving by email or picked from shared folders, then extraction, validation, matching, coding, exception handling, and posting to ERP.

Out of scope: E-invoicing channel integrations and tax-authority compliance/clearance integrations are deliberately excluded, per your instruction. So is structured-format ingestion (EDI, XML, cXML) as a primary channel.

How to read the tier column

Core. Table stakes. Build these first; the product is not credible without them.

Smart. The AI/ML differentiators. This is where the product earns its place against rules-only competitors.

Adv. Stretch or phase-2. High value, heavier lift, or dependent on data volume and maturity.

Tiering is my read of build order and differentiation, not a hard roadmap. Adjust per your engineering capacity and the first client's priorities.

Contents

1. Ingestion

How invoices enter the product. Scoped to PDF/image arriving by email or shared folder, plus manual upload as a fallback. No e-invoicing or EDI channels.

ID	Feature	What it does	Tier
IN-01	Email channel ingestion	Poll one or more shared AP mailboxes; pull PDF and image attachments, log sender, subject, received timestamp.	Core
IN-02	Shared-folder ingestion	Watch network/cloud share folders (SharePoint, network drive, S3-style bucket); pick up new PDF/image files on a schedule or on drop.	Core
IN-03	Manual upload	Drag-and-drop or browse upload for one-off invoices and batches, same pipeline as automated channels.	Core
IN-04	Format support	PDF (native and scanned), TIFF, JPEG, PNG; multi-page documents; barcode and QR code regions.	Core
IN-05	Email body capture	Read invoice data from the email body itself, not just attachments, for suppliers who paste invoices inline.	Smart
IN-06	Document classification	Separate invoices from non-invoices (statements, reminders, marketing) and route or quarantine accordingly before extraction.	Smart
IN-07	Multi-invoice splitting	Detect and split a single PDF that contains several invoices into separate work items.	Smart
IN-08	Channel reconciliation	Daily count of received vs. ingested vs. failed per channel, with a quarantine queue for unreadable files.	Core
IN-09	Duplicate-at-intake check	Hash and fingerprint files at ingestion to stop the same file entering twice from email plus folder.	Smart

2. Extraction

Turning a document image into structured invoice data. This is the Capture-side intelligence: OCR plus contextual extraction that understands document structure.

ID	Feature	What it does	Tier
EX-01	Header extraction	Pull vendor, invoice number, invoice date, due date, PO number, currency, totals, tax amounts.	Core
EX-02	Line-item extraction	Capture line description, quantity, UoM, unit price, line amount, tax per line; handle multi-page line tables.	Core
EX-03	Multi-PO extraction	Detect and capture more than one PO reference on a single invoice and tag lines to the right PO.	Smart
EX-04	Key-value-pair extraction	Configurable extraction of any labelled field beyond the standard set (e.g. delivery note, contract ref).	Smart

ID	Feature	What it does	Tier
EX-05	Handwritten / vertical text	Read handwritten annotations and rotated or vertical text blocks.	Smart
EX-06	Language detection	Detect document language and apply the right extraction model; flag mixed-language documents.	Smart
EX-07	Confidence scoring	Field-level confidence scores that drive the validation queue and skip-validation logic.	Smart
EX-08	In-image search & validation	Click a field in the UI and jump to the exact region in the source image for verification.	Smart
EX-09	Auto-retraining	Retrain the extraction engine on a client's own corrected volumes to lift accuracy over time.	Adv
EX-10	Address parsing & normalization	Split and standardize vendor/remit-to addresses into structured components for matching.	Adv
EX-11	Signature & stamp detection	Detect presence/absence of signatures and stamps where approval evidence is needed.	Adv
EX-12	Regex-driven extraction	Client-defined regex patterns for non-standard fields and reference formats.	Adv

3. Validation and enrichment

Deciding what a human needs to touch, and cleaning data before it moves downstream. The goal is to keep humans on exceptions, not on every invoice.

ID	Feature	What it does	Tier
VL-01	Confidence-based validation queue	Route only low-confidence or failed-rule documents to a human; auto-pass the rest.	Smart
VL-02	Skip-validation rules	Configurable thresholds where high-confidence invoices bypass manual review entirely.	Smart
VL-03	Post-processing business rules	Apply client rules after extraction (default tax code, derive due date, normalize vendor name).	Core
VL-04	Dynamic UI rules	Show/hide/require fields in the validation screen based on document type, country, or vendor.	Core
VL-05	Mandatory-field enforcement	Block release of any invoice missing required fields for the target ERP.	Core
VL-06	Vendor master lookup	Match extracted vendor against the vendor master; suggest the record or flag a new-vendor case.	Smart
VL-07	QC queue management	Supervisor view to assign, reassign, and track the validation backlog with SLA timers.	Adv
VL-08	Redaction	Mask sensitive fields (bank details, personal data) in stored images per client policy.	Adv

4. Matching

Reconciling the invoice against PO, goods receipt, and delivery schedules. The heart of touchless processing for PO-backed spend.

ID	Feature	What it does	Tier
MT-01	2-way / 3-way matching	Match invoice to PO (2-way) and to PO plus goods receipt (3-way) at header and line level.	Core
MT-02	GR-based matching	Match against goods-receipt quantity and value, including partial and multiple GRs.	Core
MT-03	PO line tagging (semantic)	LLM-based matching of invoice line descriptions to PO lines, including fuzzy and partial matches.	Smart
MT-04	Delivery-schedule matching	Match lines to delivery schedules and resolve between-schedule cases with a signal hierarchy.	Smart
MT-05	Tolerance logic	Configurable price/quantity tolerances at line and header; auto-accept within band, route outside.	Core
MT-06	UoM matching & conversion	Handle non-standard UoM codes and conversion-rate mismatches between invoice and PO.	Smart
MT-07	Freight / DCGR handling	Apportion planned vs. unplanned delivery costs across lines; tolerance and buyer routing for freight.	Smart
MT-08	GR variance handling	Detect and auto-resolve quantity/value variances against goods receipt where rules permit.	Smart

5. Coding

Deriving GL, tax, and dimension coding, with the heaviest lift on non-PO invoices where there is no PO to match against.

ID	Feature	What it does	Tier
CD-01	Non-PO GL coding	ML/LLM-suggested GL code, cost centre, and dimensions for invoices with no PO reference.	Smart
CD-02	Rules-based tax coding	Derive tax code from country, vendor, and line attributes using a configurable rule set.	Core
CD-03	Pre-approval determination	Decide whether a non-PO invoice needs pre-approval based on amount, category, and policy.	Smart
CD-04	Multi-dimension code selection	Suggest and validate combinations across GL, cost centre, profit centre, project, and internal order.	Smart
CD-05	Line-level mixed-state handling	Handle invoices where some lines are PO-backed and others are non-PO in a single document.	Smart
CD-06	Currency conversion	Apply the right exchange rate and post in document and local currency.	Core

6. Exception handling and routing

What happens when an invoice cannot go straight through. Configuration-driven routing, fallback cascades, and AI-suggested resolutions.

ID	Feature	What it does	Tier
EXC-01	Exception routing engine	Configuration-driven matrix that decides who resolves each exception by company code, org, doc type, etc.	Core
EXC-02	Per-row fallback cascade	Each routing row carries its own ordered fallback chain (e.g. Buyer to PO Creator to Super User).	Smart
EXC-03	SPOC assignee types	Support work-centre, warehouse, and other SPOCs as distinct assignee types in the cascade.	Smart
EXC-04	Time-based fallback & escalation	Reminders and escalation when an assignee does not act within the SLA window.	Smart
EXC-05	Auto-routing of exceptions	Automatically push matching, validation, and price exceptions to the right resolver with context.	Smart
EXC-06	Next-best-action recommendation	Suggest the likely resolution for each exception so the human starts from a pre-triaged work item.	Smart
EXC-07	Duplicate detection	Catch duplicates across vendor, amount, date, and reference variations, with similarity scores.	Smart
EXC-08	Anomaly / data-error scoring	Score invoices for data anomalies and vendor-master inconsistencies before posting.	Adv

7. Intelligence and orchestration

The agentic layer that prioritizes work, allocates it, and coordinates resolution across the pipeline. This is where the reimagined suite goes beyond a workflow tool.

ID	Feature	What it does	Tier
AG-01	AI prioritization	Rank the queue so high-value, at-risk, and time-sensitive invoices surface first.	Smart
AG-02	Work allocation	Allocate invoices to processors by skill, load, language, or client rule.	Smart
AG-03	Agentic processing crew	A crew of agents (retrieval, validation, response, monitoring, supervisor) that coordinate across the pipeline.	Adv
AG-04	Proactive assistance engine	Monitors invoice state and surfaces guidance without being asked, at configurable trigger points.	Adv
AG-05	Confidence-gated auto-execute	Auto-execute actions above a high confidence threshold; fall back to human below it.	Adv
AG-06	Human-in-the-loop checkpoints	No state change without confirmation below threshold; full audit of every agent action.	Core

8. Platform and foundations

The non-negotiable plumbing: ERP integration, audit, reporting, configuration, access control, and notifications. Unglamorous, but the product fails without it.

ID	Feature	What it does	Tier
PF-01	ERP integration layer	API-mediated read/write to SAP, Oracle Fusion, and PeopleSoft for PO, GR, vendor, and posting.	Core
PF-02	Audit trail	Immutable log of every extraction, edit, routing, and posting action per invoice.	Core
PF-03	Accuracy & reconciliation reporting	Field-level accuracy, straight-through rate, exception mix, and channel reconciliation dashboards.	Core
PF-04	Configurable workflow	No-code configuration of channels, rules, tolerances, routing, and thresholds per client.	Core
PF-05	Role-based access control	Scope every action and view to role and entity; enforce segregation of duties.	Core
PF-06	Notifications	Email/in-app alerts for assignments, escalations, and SLA breaches.	Core

Scope boundary

To keep the build focused, the following are explicitly NOT in scope for this product:

- E-invoicing channel ingestion (clearance and post-audit models) and any tax-authority connectivity.
- Tax compliance determination beyond simple rules-based tax coding from existing master data.
- Structured-format channels as primary intake: EDI, XML, cXML, supplier-portal feeds.
- Payment execution and bank connectivity. The product ends at a posted, payment-ready invoice.
- Supplier/buyer self-service portals and helpdesk query resolution (that is the Assist product's territory).
- Standalone audit/forensics tooling beyond the in-product audit trail (that is Trace's territory).

Suggested build sequence

Phase 1, prove the pipeline. IN-01 to IN-04, IN-08; EX-01, EX-02, EX-07; VL-03 to VL-05; MT-01, MT-05; CD-02, CD-06; EXC-01, EXC-06; PF-01 to PF-05. This is a working email-to-ERP path for PO-backed invoices.

Phase 2, add the intelligence. EX-03 to EX-08; VL-01, VL-02, VL-06; MT-03, MT-04, MT-06 to MT-08; CD-01, CD-03 to CD-05; EXC-02 to EXC-05, EXC-07; AG-01, AG-02. This is where straight-through rate climbs.

Phase 3, agentic and stretch. EX-09 to EX-12; VL-07, VL-08; EXC-08; AG-03 to AG-06. Higher lift, dependent on data maturity and volume.

If you want, I can turn any single row into a full feature spec (Summary / Business Rules / Edge Cases / Engineering Notes) or build this as an Excel feature list with priority, phase, and disruption scoring like the AP Advance workbook.